

CUMMINS, Australia

WMO: 956630

Lat: 34.2525S

Lon: 135.7136E

Elev: 199

StdP: 14.59

Time Zone: 9.50 (AUA)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.7	36.9	27.9	22.0	83.2	31.3	25.8	71.9	28.6	54.1	26.3	55.1	2.4	40	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.5	101.7	65.4	96.7	64.8	92.1	63.9	70.2	84.6	68.4	84.5	66.9	83.6	14.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.0	72.4	64.1	90.2	70.6	61.9	83.3	69.0	34.3	84.2	32.8	84.8	31.5	83.1	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	22.3	20.1	DB	31.1	109.5	2.5	3.5	29.3	112.0	27.8	114.0	26.4	116.0	24.6	118.5	(4)
(5)			WB	31.0	73.0	2.6	1.6	29.1	74.2	27.5	75.1	26.1	76.0	24.1	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	72.2	71.1	67.8	63.0	57.2	52.1	50.8	51.1	54.1	59.8	64.8	68.3		
	DBStd	9.99	8.11	7.72	7.46	6.19	4.60	3.52	3.49	3.87	5.31	7.54	8.69	8.33		
	HDD50	95	0	0	0	0	1	19	31	30	12	2	0	0		
	HDD65	2389	12	15	49	110	248	386	439	430	331	203	111	55		
	CDD50	4103	688	592	553	391	224	82	57	65	134	307	443	567		
	CDD65	924	235	187	137	51	6	0	0	0	3	43	105	157		
	CDH74	10980	2847	1992	1387	498	52	0	0	1	66	673	1402	2062		
	CDH80	5580	1591	1034	644	167	6	0	0	0	10	277	727	1124		
(14) Wind	WSAvg	9.9	12.1	11.5	10.5	8.9	8.6	8.1	9.2	8.7	9.0	10.0	10.7	11.3		
(15) Precipitation	PrecAvg	18.2	0.5	0.7	0.7	1.2	1.9	3.3	2.7	2.4	1.8	1.2	0.9	0.9		
	PrecMax	30.6	1.8	3.1	2.2	3.1	3.3	6.1	4.7	4.9	3.8	3.0	1.9	5.0		
	PrecMin	11.5	0.0	0.0	0.0	0.0	0.4	1.7	1.3	0.4	0.4	0.0	0.1	0.0		
	PrecStd	4.1	0.4	0.7	0.6	0.8	0.8	1.2	0.9	1.1	0.9	0.8	0.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.8	104.8	101.3	91.0	79.4	67.6	67.5	71.1	80.0	92.7	102.8	104.7		
		MCWB	66.5	67.1	64.2	63.1	59.4	55.5	55.4	58.2	59.7	60.8	64.0	65.7		
	2%	DB	102.2	97.4	92.8	84.8	74.1	64.4	62.8	66.6	75.1	87.5	94.8	98.4		
		MCWB	66.4	65.8	64.1	60.6	57.5	54.9	55.2	56.4	58.4	60.1	62.4	64.1		
	5%	DB	95.8	92.5	87.3	79.7	69.0	62.1	60.2	62.8	70.4	81.7	88.8	92.3		
		MCWB	66.1	65.0	64.0	59.3	56.5	55.1	54.5	55.3	58.0	59.5	61.4	63.1		
	10%	DB	88.8	86.1	81.3	75.0	65.7	59.9	58.2	59.8	65.4	75.5	82.1	85.8		
		MCWB	64.6	64.4	63.4	59.2	56.2	54.1	53.5	54.1	56.0	58.6	60.4	62.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.7	72.4	72.0	66.1	62.0	59.3	59.4	60.4	63.6	65.2	67.0	70.3		
		MCDB	91.5	87.1	81.0	77.9	69.1	61.5	61.7	65.3	74.2	82.0	85.9	81.7		
	2%	WB	69.9	69.8	69.1	63.7	60.3	57.8	57.3	58.2	60.7	62.9	64.8	67.8		
		MCDB	90.4	81.1	78.8	74.0	66.8	60.9	60.1	63.8	71.2	78.2	86.3	85.7		
	5%	WB	68.1	67.9	66.9	62.3	59.1	56.5	55.8	56.3	58.6	61.1	63.3	65.8		
		MCDB	87.7	84.4	79.6	71.5	65.4	60.1	58.9	61.0	66.8	75.7	82.5	83.9		
	10%	WB	66.2	66.0	65.1	61.1	57.8	55.2	54.3	54.7	56.7	59.3	61.6	63.8		
		MCDB	84.6	82.6	77.7	70.4	63.9	58.9	57.4	58.9	64.2	72.5	77.5	80.6		
(35) Mean Daily Temperature Range	MDBR		24.5	22.1	21.9	21.7	17.7	16.6	15.9	18.3	21.8	26.5	25.2	25.4		
	5% DB	MCDBR	34.6	32.1	30.0	28.1	23.4	19.8	18.1	23.0	30.0	36.0	34.8	35.4		
		MCWBR	9.8	9.0	9.3	10.5	11.1	12.6	12.1	14.5	15.7	14.3	10.8	10.9		
	5% WB	MCDBR	28.1	24.0	22.5	22.0	17.6	16.2	16.0	20.9	26.3	29.2	27.7	27.9		
		MCWBR	10.0	9.0	8.9	10.2	10.2	11.8	12.3	14.5	15.1	13.6	10.2	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.343	0.342	0.324	0.319	0.310	0.301	0.298	0.305	0.333	0.338	0.342	0.334		
	taud		2.521	2.543	2.582	2.584	2.596	2.608	2.611	2.570	2.475	2.465	2.476	2.530		
	Ebn at Noon		315	309	303	287	273	267	275	288	295	306	313	319		
	Edh at Noon		36	33	30	26	23	22	23	26	33	36	37	35		
(44) All-Sky Solar Radiation	RadAvg		2435	2142	1738	1266	891	746	815	1102	1505	1922	2232	2401		
	RadStd		108	109	78	78	40	36	36	68	104	128	125	107		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (13 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	-184	+218	N/A	(47)

Nomenclature: See separate page